

Shoplytics Documentation Report

AI-Powered Product Intelligence & Smart Shopping Copilot

1. Introduction

In the modern digital marketplace, users often struggle with comparing products across multiple platforms due to fragmented information, varying prices, and inconsistent reviews. This process is time-consuming and inefficient.

Shoplytics is developed as an intelligent solution to automate this workflow. It is a **multi-agent AI-based system** that performs real-time product research, comparison, and recommendation by interacting directly with live e-commerce websites.

The system combines **browser automation, artificial intelligence, and structured data processing** to deliver actionable insights and optimal product recommendations.

2. Objective

The primary objectives of Shoplytics are:

- To automate product comparison across multiple platforms
- To provide real-time pricing and product insights
- To analyze customer sentiment from reviews
- To detect best deals and overpriced products
- To generate explainable AI-based recommendations

3. System Overview

Shoplytics operates as a **multi-agent pipeline system**, where each agent is responsible for a specific task.

The system workflow follows:

User Query → Planning → Search → Scraping → Extraction → Deduplication → Analysis → Ranking → Recommendation → UI Display

4. Architecture

The system is built using a **modular multi-agent architecture** orchestrated via LangGraph.

4.1 Pipeline Overview

- Planner Agent
- Search Agent
- Domain Filter Agent
- Browser Agent (Playwright)
- Extraction Agent
- Deduplication Agent
- Sentiment Analysis Agent
- Deal Detection Agent
- Ranking Agent
- Summary Agent

4.2 Agent Responsibilities

4.2.1 Planner Agent

Interprets the user query and creates a structured execution plan.

4.2.2 Search Agent

Fetches relevant product URLs using search engines.

4.2.3 Domain Filter Agent

Filters results to trusted e-commerce platforms such as Amazon, Flipkart, and Croma.

4.2.4 Browser Agent

Uses Playwright to open websites and retrieve dynamic content.

4.2.5 Extraction Agent

Extracts structured data (name, price, rating, etc.) from HTML content.

4.2.6 Deduplication Agent

Removes duplicate product listings across platforms.

4.2.7 Sentiment Analysis Agent

Analyzes customer reviews to determine overall sentiment.

4.2.8 Deal Detection Agent

Evaluates whether a product is underpriced, fairly priced, or overpriced.

4.2.9 Ranking Agent

Assigns a score to each product based on weighted factors.

4.2.10 Summary Agent

Generates a final recommendation with reasoning.

5. Technology Stack

Component	Technology
Backend	Python
Frontend	Streamlit
Browser Automation	Playwright
Agent Orchestration	LangGraph
AI Models	OpenAI / Gemini / Groq
Visualization	Plotly
Concurrency	Asyncio

6. Working Methodology

Step 1: User Input

User provides a natural language query.

Step 2: Planning

Planner agent breaks the query into subtasks.

Step 3: Search

Search agent retrieves relevant product links.

Step 4: Scraping

Browser agent opens websites and loads content dynamically.

Step 5: Extraction

Extractor converts raw HTML into structured data.

Step 6: Deduplication

Duplicate entries are removed.

Step 7: Sentiment Analysis

Reviews are analyzed to determine positivity score.

Step 8: Deal Detection

Prices are compared with market averages.

Step 9: Ranking

Products are ranked using weighted scoring.

Step 10: Recommendation

Best product is selected and explained.

7. Ranking Algorithm

The ranking system uses a weighted scoring model:

- Price → 40%
- Rating → 30%
- Brand Reputation → 20%
- Seller Trust → 10%

Final score is calculated on a scale of 0–100.

8. User Interface

The UI is designed as a modern dashboard with three main sections:

8.1 Input Panel

Allows users to enter queries and view pipeline execution.

8.2 AI Decision Intelligence Panel

Displays reasoning behind recommendations.

8.3 Product Comparison Table

Shows structured product data including:

- Product Name
- Price
- Rating
- Store
- Deal Status
- Sentiment

- Score

9. Key Features

- Multi-agent AI architecture
- Real-time web scraping
- Product deduplication
- Sentiment analysis
- Deal detection engine
- AI-based ranking system
- Explainable recommendations
- Interactive dashboard

10. Advantages

- Eliminates manual product comparison
- Provides real-time data
- Ensures unbiased recommendations
- Improves decision-making efficiency
- Scalable multi-agent design

11. Limitations

- Dependent on website structure changes
- Some websites may block scraping
- Sentiment accuracy depends on available reviews
- Requires API keys for LLMs

12. Future Scope

- Price history tracking
- Personalized user recommendations
- Voice-based queries

- Mobile application
- Advanced deal prediction

13. Conclusion

Shoplytics demonstrates the practical application of **multi-agent AI systems in real-world decision-making tasks**.

By combining automation, data extraction, and intelligent analysis, it transforms the traditional shopping experience into an efficient and insight-driven process.

The system highlights the potential of AI agents in solving complex, multi-step problems and delivering actionable recommendations in real time.

14. References

- LangGraph Documentation
- Playwright Documentation
- Streamlit Documentation
- OpenAI / Google Gemini APIs